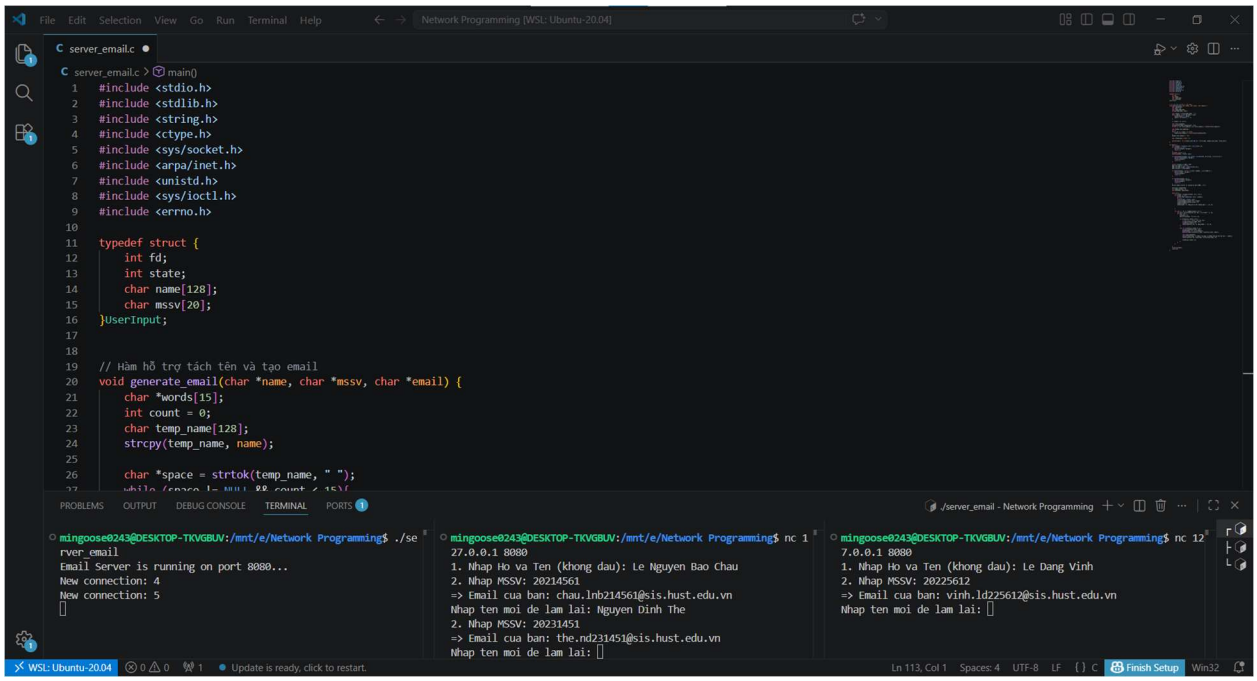
BÁO CÁO BÀI TẬP NGÀY 31/03/2026
HỌC PHẦN: LẬP TRÌNH MẠNG

Sinh viên: Hoàng Hà My
MSSV: 20235383

Bài 1: Lập trình ứng dụng server trả về địa chỉ email của sinh viên theo yêu cầu sau:

- **Khi có client kết nối, server gửi thông điệp để hỏi "Họ tên" và "MSSV"**
- **Từ dữ liệu nhận được, server phản hồi địa chỉ email cho client, theo cấu trúc email của sv ĐHBK**
- **Sử dụng non-blocking server, thử nghiệm với nhiều client kết nối cùng một lúc.**



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Network Programming [WSL: Ubuntu-20.04]

server_email.c

server_email.c > main()

```c
    while (space != NULL && count < 15){
        words[count++] = space;
        space = strtok(NULL, " ");
    }

    if (count < 1) return;

    char first_name[64];
    strcpy(first_name, words[count - 1]);
    for(int i = 0; first_name[i]; i++) first_name[i] = tolower(first_name[i]);

    char middle_last_name[15];
    int i;
    for(i = 0; i < count - 1; i++){
        middle_last_name[i] = (char)tolower(words[i][0]);
    }
    middle_last_name[i] = '\0';

    char *final_mssv = mssv + 2;

    sprintf(email, "%s.%s%s@sis.hust.edu.vn", first_name, middle_last_name, final_mssv);
}

int main() {
    int listener = socket(AF_INET, SOCK_STREAM, 0);
    if (listener == -1) {
        perror("socket() failed");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 1

./server_email - Network Programming

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./se
rver_email
Email Server is running on port 8080...
New connection: 4
New connection: 5
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 1
27.0.0.1 8080
1. Nhap Ho va Ten (khong dau): Le Nguyen Bao Chau
2. Nhap MSSV: 20214561
=> Email cua ban: chau.lnb214561@sis.hust.edu.vn
Nhap ten moi de lam lai: Nguyen Dinh The
2. Nhap MSSV: 20231451
=> Email cua ban: the.nd231451@sis.hust.edu.vn
Nhap ten moi de lam lai:
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 12
7.0.0.1 8080
1. Nhap Ho va Ten (khong dau): Le Dang Vinh
2. Nhap MSSV: 20225612
=> Email cua ban: vinh.ld225612@sis.hust.edu.vn
Nhap ten moi de lam lai:
```

WSL: Ubuntu-20.04 ⊗ 0 ⚠ 0 1 Update is ready, click to restart. Ln 113, Col 1 Spaces: 4 UTF-8 LF {} C Finish Setup Win32

File Edit Selection View Go Run Terminal Help

Network Programming [WSL: Ubuntu-20.04]

server_email.c

server_email.c > main()

```c
    int listener = socket(AF_INET, SOCK_STREAM, 0);
    if (listener == -1) {
        perror("socket() failed");
        return 1;
    }
    unsigned long ul = 1;
    ioctl(listener, FIONBIO, &ul);

    if (setsockopt(listener, SOL_SOCKET, SO_REUSEADDR, &(int){1}, sizeof(int))) {
        perror("setsockopt() failed");
        close(listener);
        return 1;
    }

    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    addr.sin_addr.s_addr = htonl(INADDR_ANY);
    addr.sin_port = htons(8080);

    if (bind(listener, (struct sockaddr *)&addr, sizeof(addr))) {
        perror("bind() failed");
        close(listener);
        return 1;
    }

    if (listen(listener, 5)) {
        perror("listen() failed");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 1

./server_email - Network Programming

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./se
rver_email
Email Server is running on port 8080...
New connection: 4
New connection: 5
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```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 1
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Nhap ten moi de lam lai:
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 12
7.0.0.1 8080
1. Nhap Ho va Ten (khong dau): Le Dang Vinh
2. Nhap MSSV: 20225612
=> Email cua ban: vinh.ld225612@sis.hust.edu.vn
Nhap ten moi de lam lai:
```

WSL: Ubuntu-20.04 ⊗ 0 ⚠ 0 1 Update is ready, click to restart. Ln 113, Col 1 Spaces: 4 UTF-8 LF {} C Finish Setup Win32

File Edit Selection View Go Run Terminal Help — Network Programming [WSL: Ubuntu-20.04]

server_email.c > main()

```c
        perror("listen() failed");
        close(listener);
        return 1;
    }
    printf("Email Server is running on port 8080...\n");

    UserInput clients[64];
    int numberclients = 0;
    char buf[256], email[256];

    while (1) {
        int client = accept(listener, NULL, NULL);
        if (client != -1) {
            printf("New connection: %d\n", client);
            ul = 1;
            ioctl(client, FIONBIO, &ul);
            clients[numberclients].fd = client;
            clients[numberclients].state = 0;
            numberclients++;
            send(client, "1. Nhap Ho va Ten (khong dau): ", 31, 0);


        }

        for (int i = 0; i < numberclients; i++) {
            int len = recv(clients[i].fd, buf, sizeof(buf) - 1, 0);
            if (len > 0) {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 1 — ./server_email - Network Programming

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./server_email
Email Server is running on port 8080...
New connection: 4
New connection: 5
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 127.0.0.1 8080
1. Nhap Ho va Ten (khong dau): Le Nguyen Bao Chau
2. Nhap MSSV: 20214561
=> Email cua ban: chau.lnb214561@sis.hust.edu.vn
Nhap ten moi de lam lai: Nguyen Dinh The
2. Nhap MSSV: 20231451
=> Email cua ban: the.nd231451@sis.hust.edu.vn
Nhap ten moi de lam lai:
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 127.0.0.1 8080
1. Nhap Ho va Ten (khong dau): Le Dang Vinh
2. Nhap MSSV: 20225612
=> Email cua ban: vinh.ld225612@sis.hust.edu.vn
Nhap ten moi de lam lai:
```

WSL: Ubuntu-20.04 — Update is ready, click to restart. — Ln 113, Col 1 — Spaces: 4 — UTF-8 — LF — C — Finish Setup — Win32

File Edit Selection View Go Run Terminal Help — Network Programming [WSL: Ubuntu-20.04]

server_email.c > main()

```c
            buf[len] = 0;
            buf[strcspn(buf, "\r\n")] = 0;

            if (clients[i].state == 0) {
                // Luu tên và chuyển sang đợi MSSV
                strcpy(clients[i].name, buf);
                clients[i].state = 1;
                send(clients[i].fd, "2. Nhap MSSV: ", 14, 0);
            }

            else if (clients[i].state == 1) {
                strcpy(clients[i].mssv, buf);
                memset(email, 0, sizeof(email));
                generate_email(clients[i].name, clients[i].mssv, email);

                char final_msg[512];
                sprintf(final_msg, "=> Email cua ban: %s\nNhap ten moi de lam lai: ", email);
                send(clients[i].fd, final_msg, strlen(final_msg), 0);

                clients[i].state = 0;
            }
        }
    }

    }
    close(listener);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 1 — ./server_email - Network Programming

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./server_email
Email Server is running on port 8080...
New connection: 4
New connection: 5
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 127.0.0.1 8080
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2. Nhap MSSV: 20231451
=> Email cua ban: the.nd231451@sis.hust.edu.vn
Nhap ten moi de lam lai:
```
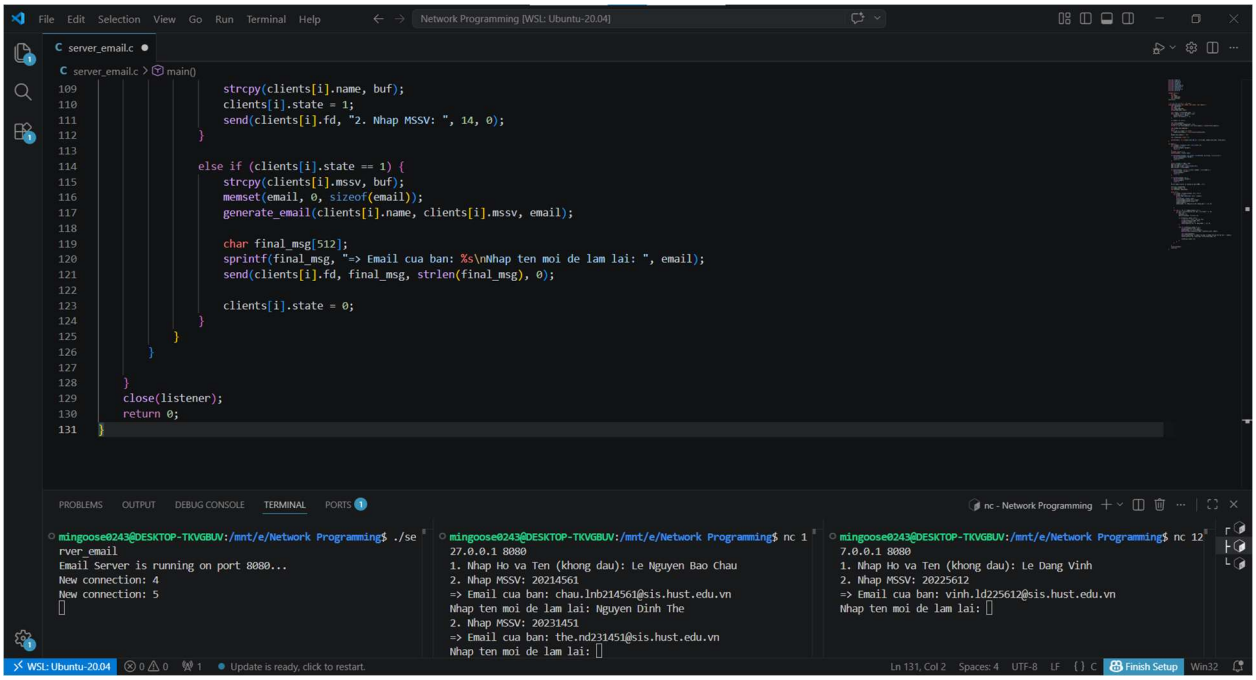
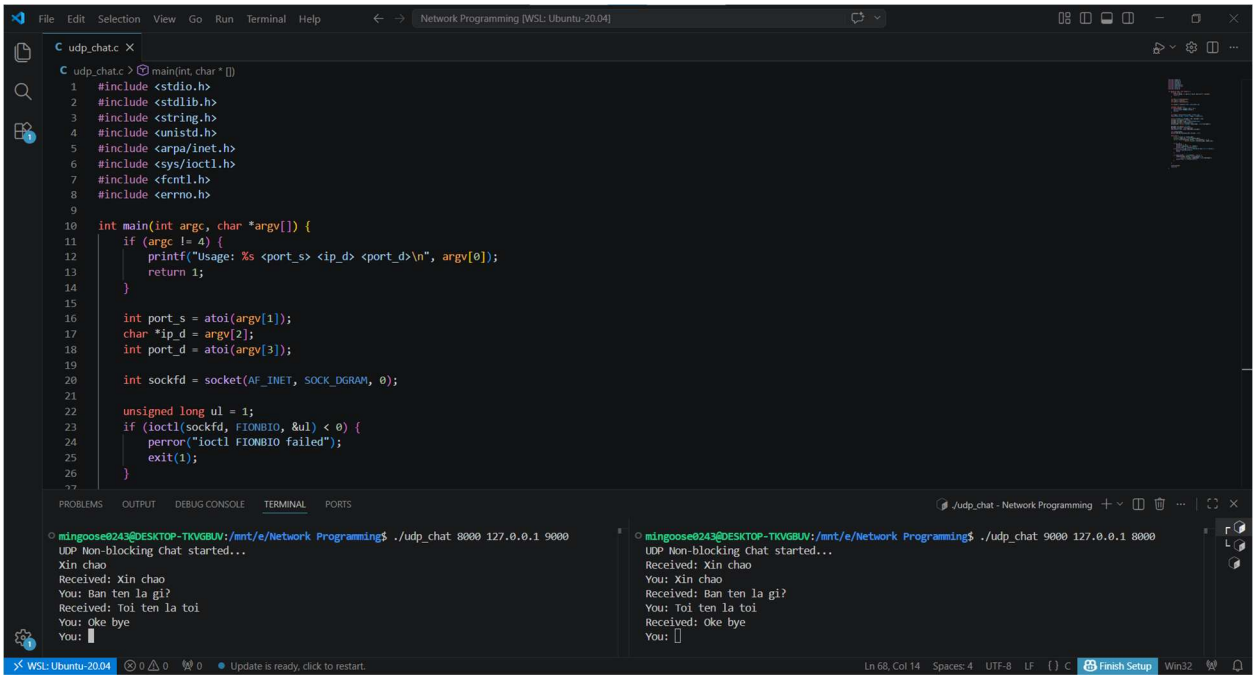

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ nc 127.0.0.1 8080
1. Nhap Ho va Ten (khong dau): Le Dang Vinh
2. Nhap MSSV: 20225612
=> Email cua ban: vinh.ld225612@sis.hust.edu.vn
Nhap ten moi de lam lai:
```

WSL: Ubuntu-20.04 — Update is ready, click to restart. — Ln 129, Col 21 — Spaces: 4 — UTF-8 — LF — C — Finish Setup — Win32



Bài 2: Lập trình ứng dụng UDP chat sử dụng kỹ thuật non-blocking, cho phép vừa gửi và nhận dữ liệu trong một ứng dụng. Ứng dụng nhận 3 tham số dòng lệnh: udp_chat port_s ip_d port_d. Trong đó port_s là cổng nhận dữ liệu trên ứng dụng hiện tại, ip_d và port_d là địa chỉ IP và cổng của ứng dụng cần kết nối



udp_chat.c > main(int, char * []*)

```c
    int flags = fcntl(STDIN_FILENO, F_GETFL, 0);
    fcntl(STDIN_FILENO, F_SETFL, flags | O_NONBLOCK);

    struct sockaddr_in servaddr = {0}, destaddr = {0};
    servaddr.sin_family = AF_INET;
    servaddr.sin_addr.s_addr = htonl(INADDR_ANY);
    servaddr.sin_port = htons(port_s);
    bind(sockfd, (struct sockaddr *)&servaddr, sizeof(servaddr));

    destaddr.sin_family = AF_INET;
    destaddr.sin_port = htons(port_d);
    inet_pton(AF_INET, ip_d, &destaddr.sin_addr);

    char buffer[1024];
    printf("UDP Non-blocking Chat started...\n");

    while (1) {
        struct sockaddr_in remote_addr;
        socklen_t addr_len = sizeof(remote_addr);
        int n = recvfrom(sockfd, buffer, sizeof(buffer) - 1, 0,
                         (struct sockaddr *)&remote_addr, &addr_len);

        if (n > 0) {
            buffer[n] = '\0';
            printf("\rReceived: %s", buffer);
            printf("You: "); fflush(stdout);
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./udp_chat 8000 127.0.0.1 9000
UDP Non-blocking Chat started...
Xin chao
Received: Xin chao
You: Ban ten la gi?
Received: Toi ten la toi
You: Oke bye
You:
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./udp_chat 9000 127.0.0.1 8000
UDP Non-blocking Chat started...
Received: Xin chao
You: Xin chao
Received: Ban ten la gi?
You: Toi ten la toi
Received: Oke bye
You:
```

udp_chat.c > main(int, char * []*)

```c
                         (struct sockaddr *)&remote_addr, &addr_len);

        if (n > 0) {
            buffer[n] = '\0';
            printf("\rReceived: %s", buffer);
            printf("You: "); fflush(stdout);
        } else if (n < 0 && errno != EWOULDBLOCK && errno != EAGAIN) {
            perror("recvfrom error");
            break;
        }

        if (fgets(buffer, sizeof(buffer), stdin)) {
            sendto(sockfd, buffer, strlen(buffer), 0,
                   (struct sockaddr *)&destaddr, sizeof(destaddr));
            printf("You: "); fflush(stdout);
        }

    }

    close(sockfd);
    return 0;
}
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./udp_chat 8000 127.0.0.1 9000
UDP Non-blocking Chat started...
Xin chao
Received: Xin chao
You: Ban ten la gi?
Received: Toi ten la toi
You: Oke bye
You:
```

```
mingoose0243@DESKTOP-TKVGBUV:/mnt/e/Network Programming$ ./udp_chat 9000 127.0.0.1 8000
UDP Non-blocking Chat started...
Received: Xin chao
You: Xin chao
Received: Ban ten la gi?
You: Toi ten la toi
Received: Oke bye
You:
```